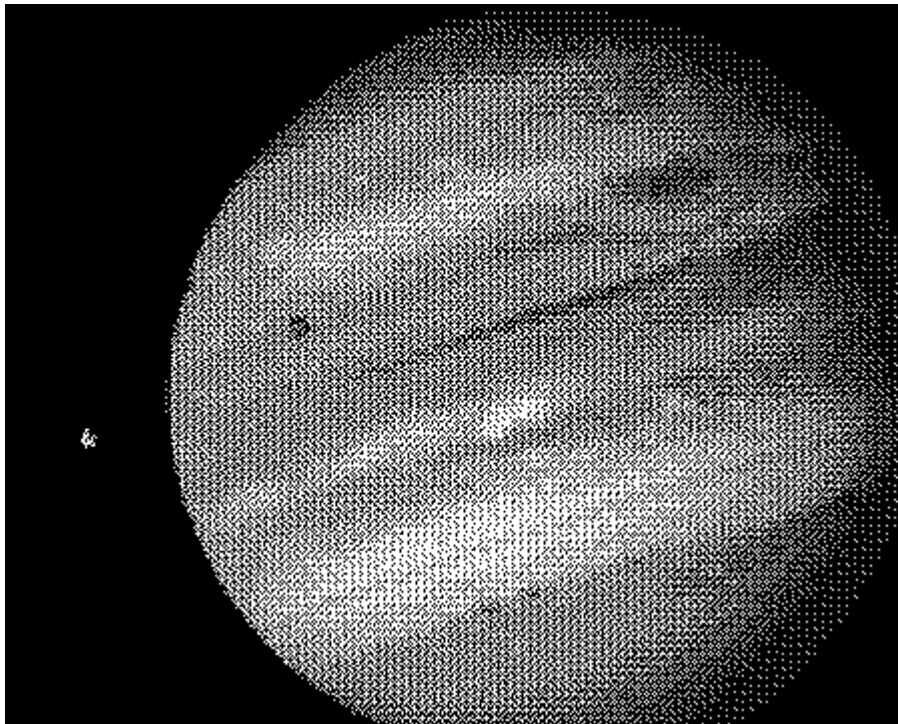


THE MOONS OF JUPITER for ASTRONOMY 121/124

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1 Abstract

The purpose of this lab is to determine the mass of Jupiter by measuring the orbital properties of Jupiter's Galilean Moons (Io, Europa, Ganymede, & Callisto, all discovered by Galileo in the seventeenth century) and analyzing their motions using Newton's Modification of Kepler's Third Law. We make the reasonable assumption that the back and forth motion of the moons results from viewing their circular orbits edge on. Using a number of observations spanning about three weeks, we plot the position of the moon as a function of time. We draw the best fitting sine curve through this data, and this allows us to mea

sure the period and radius of the orbit of each moon. By using Kepler's Third Law to interpret these period/radius measurements, we determine the mass of Jupiter.

2 Introduction

Sometimes it is possible to deduce the properties of celestial bodies without measuring them directly. For example, it is possible to measure the mass of an object by observing the orbital motion of other objects about it. To understand why, w

e need to briefly review a little history. In 1543, Nicolaus Copernicus hypothesized that the planets revolve in circular orbits around the sun. Tycho Brahe (1546–1601) carefully observed the locations of the planets and 777 stars over a period of 20 years using a sextant and compass. These observations were used by Johannes Kepler, a student of Brahe's, to deduce three empirical mathematical laws governing the orbit of one object around another. Kepler's third law for a moon orbiting a much more massive parent body is:

$$M = \frac{a^3}{P^2}$$

where M is the mass of the primary body, in units of the solar mass ($2 \times 10^{30}\text{kg}$).

a is the length of the semi-major axis of the elliptical orbit in the units of the mean Earth-Sun distance, 1 A.U. (astronomical unit). *If the orbit is circular (as will be assumed in this lab) the semi-major axis is the same as the **radius** of the orbit.*

P is the period of the orbit in Earth years. The period is the amount of time required for the moon to orbit the parent body once.

This law could apply to planets orbiting about the Sun (check: for the Earth $a = 1$ and $P = 1$ in the given units, and for the Sun $M = 1$) or to any moon orbiting around its planet.

In 1609, Hans Lippershey, a Dutch spectacle maker, invented a new optical instrument. By using two lenses, one convex and one concave, Lippershey found that distant objects could be made to look nearer. He called this new invention a telescope. Without even having seen an assembled telescope, Galileo was able to construct his own telescope with a magnification of about three, allowing the observation of objects not visible to the naked eye. He soon perfected the design and construction of the telescope, and became famous as the

builder of the world's best telescopes. His best telescopes had a magnification of about thirty.

Galileo immediately began observing celestial objects with his crude instrument. He was a careful observer, and soon published a small book of his remarkable discoveries called the "Sidereal Messenger." One can imagine the excitement these new discoveries caused in the scientific community. Suddenly, a whole new world was opened! Galileo found sunspots on the Sun, and craters on the Moon. He found that Venus had phases, much as the Moon has phases. He was able to tell that the Milky Way was a myriad of individual stars. He could see that there was something strange about Saturn, but his small telescope was not able to resolve its rings.

One of the most important discoveries was that Jupiter had four moons revolving around it. Galileo made such exhaustive studies of these moons that they have come to be known as the Galilean satellites. This "miniature solar system" was clear evidence that the Copernican theory of a Sun-centered solar system was physically possible.

Prevailing religious and political dogma of his period compelled Galileo to neither "hold nor defend" the Copernican hypothesis. Nevertheless, in 1632 he published his Dialogue on the Great World System which was a thinly disguised defense of the Copernican system. This led to his forced denunciation of the theory and confinement to his home for the rest of his life.

In this lab, you will repeat Galileo's observations (without threat of government condemnation). Today, however, we also know the size of Jupiter. Jupiter's diameter is 11 times Earth's diameter, or about 1.5×10^5 kilometers. The data collected in this experiment and this information allow us to determine the mass of Jupiter.

3 Technique

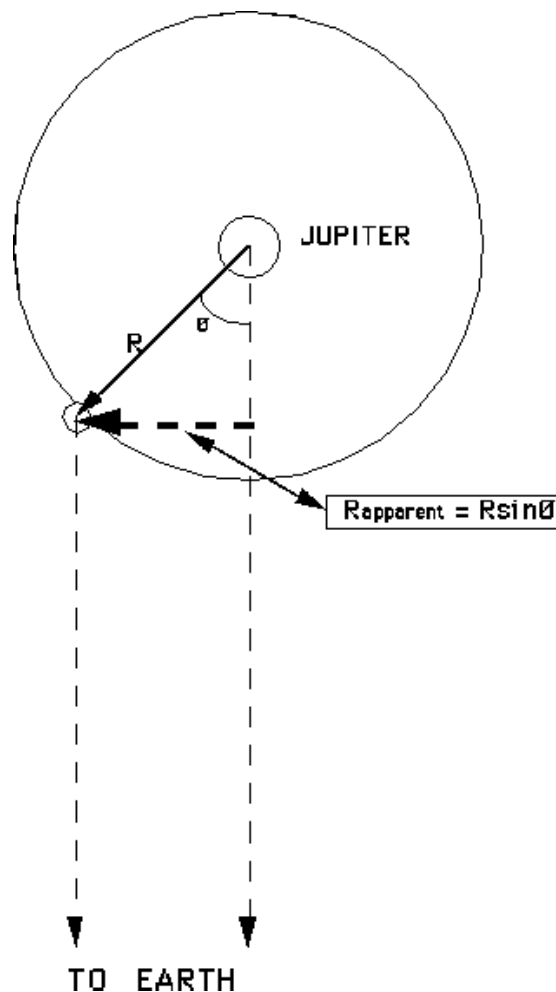
This lab can in principle be done by anyone with a set of binoculars or a small telescope. You will obtain data from at least 22 clear observing sessions making observations once per evening, spaced 24 hours apart. The computer simulation "The Moons of Jupiter" replaces actual observing sessions at the observatory using the telescope. The program simulates the operation of an automatically controlled telescope with a charge coupled device (CCD) camera that provides a video image to a computer screen. It has a sophisticated computer program that allows convenient measurements to be made at the computer console. The computer simulation is realistic in all important ways, and using it will give you a good feel for how astronomers collect data and control their telescopes. You will note that the computer also simulates some of the pitfalls of actual live telescopic observations.

We could do this lab for any one moon of Jupiter. If we did the experiment very accurately, the answer for Jupiter's mass should be the same whatever moon we used. But in practice, there are always measurement errors, which introduce inaccuracy in the determination of Jupiter's mass. Therefore we use data on all four Galilean moons of Jupiter, and by taking the average value from each moon, we improve the accuracy of our mass determination. The moons are named Io, Europa, Ganymede and Callisto, in order of distance from Jupiter. You can remember the order by the mnemonic "**I Eat Green Carrots.**" We also refer

to them in this exercise as I, II, III and IV. If you looked through any small telescope, you might see the following:



The moons would not actually appear as big as they do here, but they would appear to be lined up because we are looking edge-on to the orbital plane of the moons of Jupiter. As time goes by, the moons move around Jupiter. While the moons move in roughly circular orbits, you can only see the perpendicular distance of the moon to the line-of-sight between Jupiter and Earth.



Therefore, the perpendicular distance of the moon should be a curve if you plot it versus time. *By taking enough measurements of the position of a moon, you can fit a sine curve to the data and determine the radius of the orbit (the **amplitude of the sine curve**) in units of Jupiter diameters (J.D.) and the period of the orbit (the **period of the sine curve**) in Earth days.* Once you know the radius and period of the orbit of that moon and convert them into the appropriate units (A.U.'s and Earth years), you can determine the mass of Jupiter by using Kepler's Third Law. You will determine Jupiter's mass for each of

the four moons; there will be errors in the mass measurement when obtaining the data from the computer and when plotting and fitting the data to a sine curve, therefore your Jupiter masses may not be exactly the same.

On the computer screen, Jupiter and its moons will look much like Galileo's drawings. West is to the right and east is to the left. This is the way the sky looks through a telescope. It will be necessary to record the distance of a moon from Jupiter's center in units of Jupiter diameters (J.D.). The measurement mode of the computer equipment you will use provides a direct readout in J.D.

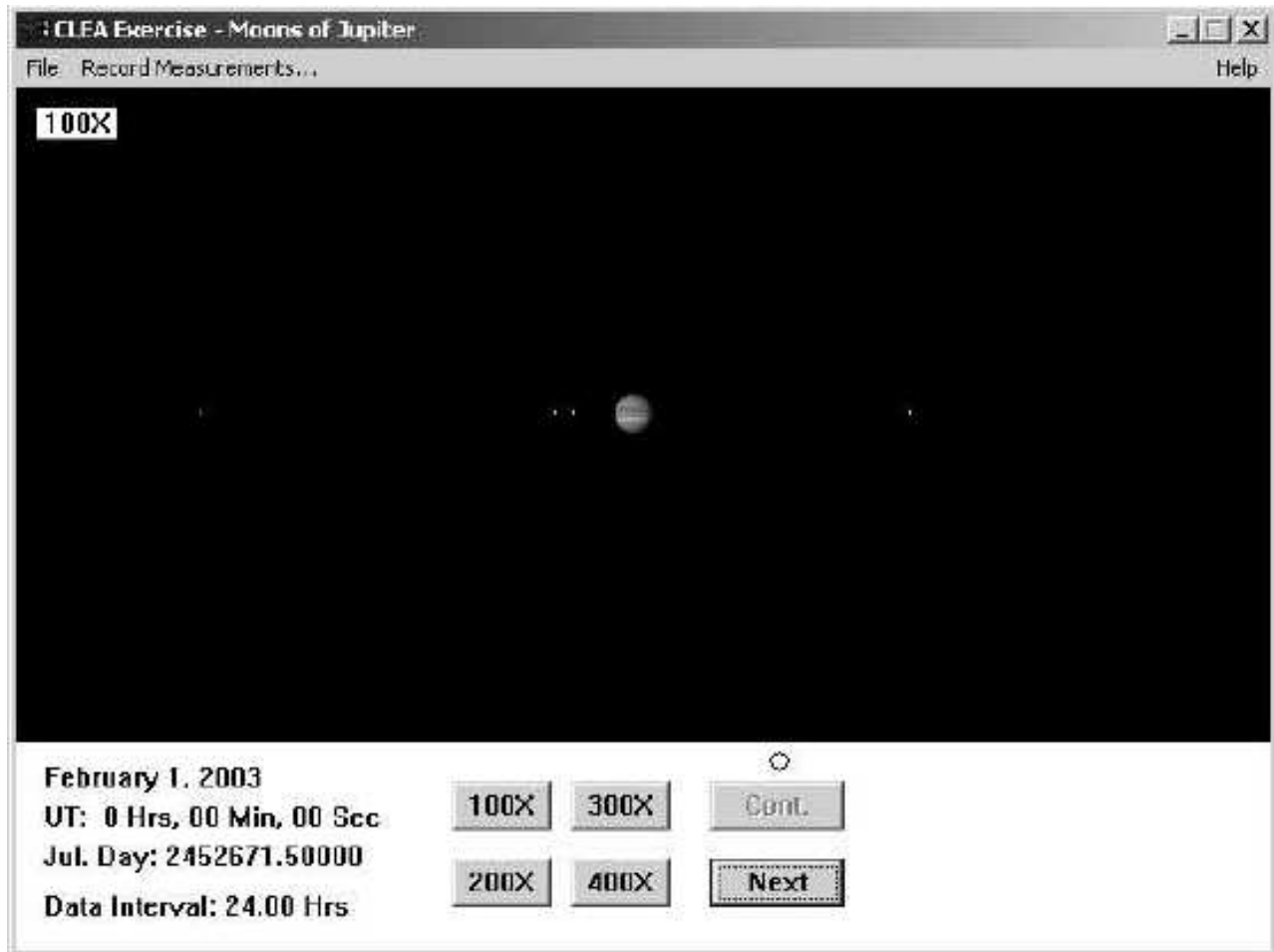
The computer simulation will be presenting data on the moons as they would be seen every 24 hours. Unfortunately, this means that you will be **undersampling** very short period moons. This is only important for Io, which has a period of roughly two days. If you wish, you may switch to 12 hour intervals to sample Io, but if you sample other moons on 12 hour intervals, you will have to make more observations (40) in order to get the period of the outer moons, e.g. Callisto. The observations are complete when you have obtained a total of at least 22 actual observations. For each moon and for each of the 22 observing sessions, you are to measure the distance of the moon from Jupiter. The data you will collect should be placed on the data sheet.

4 Procedure

- Open the Jupiter program by either double-clicking on its icon with the mouse or clicking **Start**, then selecting **Programs: CLEA Exercises: Jupiter Moons** from the menu.
- At any time, you may select **Help** from the menu to receive on-line help. Within the help menu are four choices that you can select: **On Help**, **Topics**, **User** and **About This Exercise**. **On Help** advises you on how to use the help functions within the CLEA lab. **Topics** gives detailed help on many of the commands and features of the lab. There is no user help for this program, therefore **User** is disabled. **About This Exercise** displays the title and version number of the program as well as copyright information. To close the help windows, select **Close** from the box in the lower left corner of the window. You may leave the help windows open if you prefer while working.
- You must first select **Log In...** from the **File** menu at the top left side of the screen before taking data. Enter your name on the first line. **Note: you must work individually on this laboratory.** You can leave the rest of the entries including the table number blank. Just click **OK** to continue, and when it asks **Have you finished logging in?**, click **Yes**.
- Click on **File: Run**. Set the starting date as your birthday for the current year. Assume you are observing at 9:00 pm EST, which corresponds to 02^h:00^m:00^s Universal Time (UT, i.e. the time at Greenwich, England). So enter 2^h, 0^m, 0^s for the time.

Use the Tab button or double click on each box that you wish to alter, then type the corrected value. If you wish to return to the original information, click **Cancel**. When all the information has been entered to your satisfaction, click **OK** to continue.

- After you have clicked the **OK** button, you will see a display similar to the following:



Jupiter is in the center of the screen, while the small point-like moons are to either side. You can display the screen at four scales of magnification by clicking on the **100X**, **200X**, **300X** and **400X** buttons at the bottom of the screen. In order to improve the accuracy of your measurement of a moon, you should use the largest possible magnification which leaves the moon on the screen. Even at high magnifications, the moons are very small compared to Jupiter. The current telescope magnification is displayed at the upper left hand corner of the screen, and the date and UT (the time in Greenwich) are displayed at the lower left hand corner.

- In order to measure the perpendicular distance of each moon from Jupiter, move the pointer until the tip of the arrow is centered on each moon and click the mouse. Information about the selected moon will appear at the lower right corner of the screen: the

name of the moon, the x and y pixel location on the screen, the perpendicular distance (in units of Jupiter's diameter) from the center of Jupiter along the Earth-Jupiter line-of-sight for the selected moon; and a designation of E or W to signify whether it is east or west of Jupiter. If the moon's name does not appear, you did not center the arrow exactly on the moon; try again. Occasionally a moon will pass in front of Jupiter (making it hard to find) or behind Jupiter (so it cannot be seen at all). If you can't find it, mark it down as 0.00 and go on (this is a reasonable approximation). Notice the edge on Jupiter is 0.5 J.D. To measure each moon accurately, switch to the highest magnification setting that leaves that moon on the screen. Record the data on the attached data sheet.

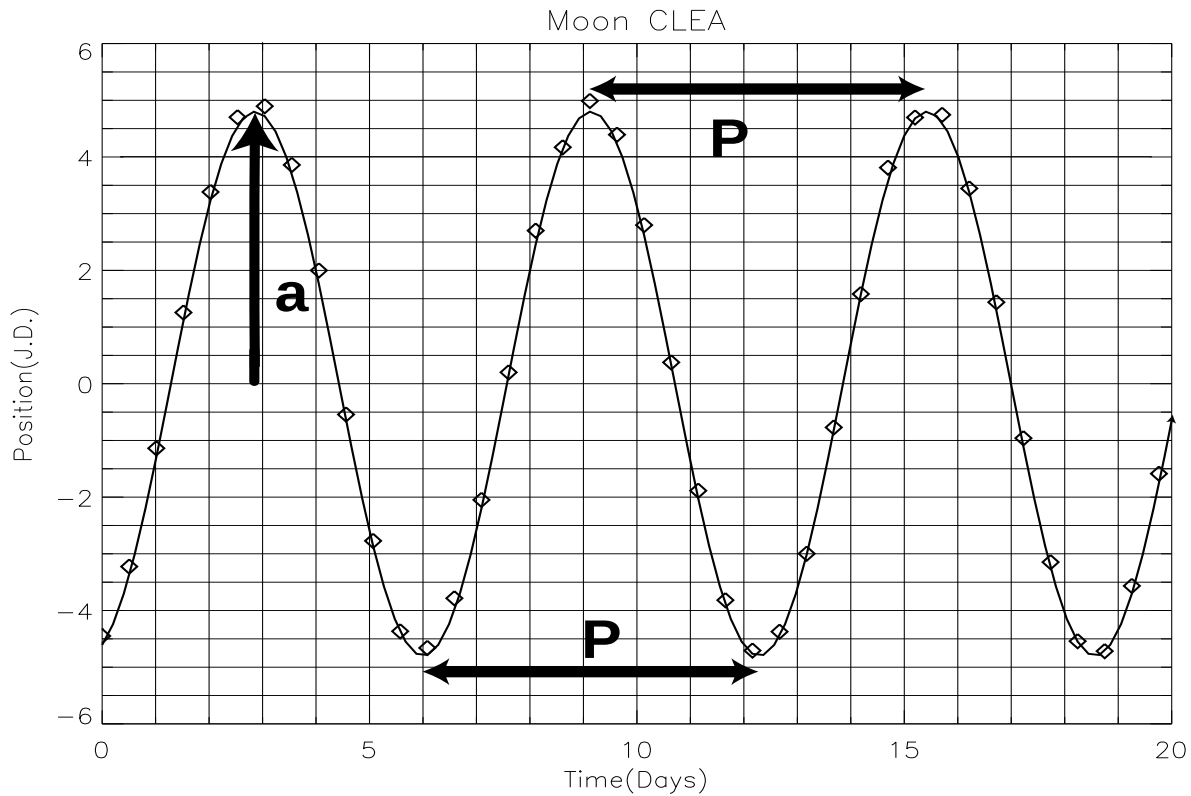
NOTE: the number given at the bottom left corner of the screen is the 'Julian Date' of observation (e.g. 'JD: 2449356.50000'). DO NOT confuse this with the measurement of the moon positions, which is given in units of Jupiter Diameters (also JD!!!). Ignore the radial distance from the moon to Jupiter (e.g. R=2.76). Record only the X-position on your table (e.g. X=3.10E).

Column 1: Local Date
 Column 2: Universal Time (02^h:00^m:00^s)
 Column 3: Day - Number of days since the first observation; remember to include cloudy days by simply writing "Cloudy" through the observations for that day
 Column 4-7: Record each moon's position under the column for that moon. **Use + for west and - for east. For example, if Europa were selected and had $X = 2.75W$, you would enter that in column 5 as +2.75.**

- When you have recorded the Universal Time (this will be the same for every observation since you are using a 24 hour interval) and the perpendicular distances for every moon, you may make the next set of observations by clicking on the **Next** button. Include cloudy days in the data set. This will allow you to get the correct spacings (leave the day blank) when plotting the motions of the moons and make it easier to fit a sine curve to it.
- When you have finished taking readings, you may quit the program by selecting **Exit** from the **File** menu.

5 Analysis

You will use the data to obtain a graph similar to the one below, by plotting position versus time (**Note:** the data shown are for an imaginary moon named CLEA, not one of the moons in the lab).



$$P = 6.2 \text{ days} = 0.0170 \text{ years}$$

$$a = 4.8 \text{ J.D.} = 0.00457 \text{ A.U.}$$

Each diamond in the figure is one observation of moon CLEA. The points are spaced at mid-day intervals, rather than full day intervals as you will do for this lab. The curve drawn through the points is the smooth curve that would be observed if you decreased the observation interval to a very small value. The shape of the curve is called a sine curve.

You will need to determine the sine curve that fits your data best in order to determine the orbital properties of each moon. Here are a few hints: **the orbits of the moons are regular (periodic)**, that is, they do not speed up or slow down from one period to the next, and **the radius of each orbit does not change** from one period to the next. The sine curve that you draw **should therefore also be regular**. You must draw a “best-fit” sine curve through each moon’s data points, such that the height of the curve and the length of the curve between cycles are constant from peak to peak. **The data points will not necessarily be located at the turning points in the sine curve. They will more than likely fall in between the turning points, and thus the peaks and troughs will have to be interpolated.**

Using the data from the imaginary moon CLEA, it is possible to determine the radius and period of the orbit. The period is the time it takes to get to the same point in the orbit again. Thus the time between two maxima, for example, is the period. The time between crossing

at 0 J.D. is equal to half of the period because this is the time it takes to get from the front of Jupiter to the back of Jupiter, or half-way around. For your moons, you may not get enough observations for a full period, so you may find the time between crossing at 0 J.D. to be of use to you in determining the period, even though the moon has not gone through a complete orbit. On the other hand if you have enough observations for several cycles, you can find a more accurate period by taking the time it takes for a moon to complete, say, 4 cycles, and then divide that time by 4. **If there are several cycles, read off the time interval that contains the largest number of cycles and then divide by the number of cycles to achieve best accuracy.** In addition, it is possible to sample the period of a moon too sparsely, in which case the pattern of the sine curve will be hard to discern. For such cases you will need to approximate the sine curve peaks, taking care to go through *every* data point.

When a moon is at the maximum position eastward or westward, it is the largest apparent distance from the planet. Remember that the orbits of the moons are nearly circular, but since we see the orbits edge on, we can only determine the radius when the moon is at its maximum position eastward or westward.

- Enter the data for each moon on the graph provided. The vertical and horizontal scales are already marked and labeled. Each day's measurement of a moon's apparent separation from Jupiter should give you one dot on the graph for that moon. On days which were cloudy, skip that day number on your graph and move on to the next.

**DO THE FOLLOWING FOR EACH MOON EXCEPT IO.
START WITH CALLISTO, THEN GANYMEDE, AND STOP WITH
EUROPA!**

- For each moon except Io draw a smooth sine curve through the points. They need not fall on one of the grid lines. The curve should be symmetric about the horizontal axis; i.e. the maxima and minima should have the same values, except for their sign.

FOR IO

- Look at the Io graph. Do the plotted points look like they form a nice sine curve like the plots of the other three moons? The reason that they do not is because Io has been undersampled. That is, its orbital period is too short for one day sampling intervals. Thus, there are not enough data points on the graph to represent the sine curve. It turns out that half-day intervals will suffice. However, half-day intervals are not realistic since the moons can only be observed at night from earth. So, in order to reveal the sine curve, more data points need to be taken in intervals that are difficult to achieve for a single observer here on earth.

1. Restart the Moons of Jupiter Lab.
2. Use the same Start Day and Universal Time that you used before.
3. This time set the observational interval to 12 hrs. rather than 24 hrs. for half-day intervals. To do so, select **File: Preferences: Timing**. In the box next to **Observation Interval**, enter **12.00**. Leave the value for **Animation Step** unchanged.
4. Retake the measurements of Io only. This time record the data directly to your plot, plotting only those points which are new with a different symbol than what was used for the old data. Remember that these new points will fall mid-way between successive days, on half-day values.
5. Make sure you record all the way up to your previous ending date.
6. Now fit a curve to all of the data points. It should be easier to make out the sine wave function.

DO THE FOLLOWING FOR ALL MOONS

- Determine the period, P , and the orbital radius, a , from your graphs, as described on pages 8 and 9. **Record the value found for the period to one tenth (0.1) of a day and for the radius to one tenth of a J.D.** These values will have units of days for P and J.D. for a . In order to use Kepler's Third Law, you need to convert the period into years by dividing by the number of days in a year (365.25 days per year) and the orbital radius to A.U. by dividing by the number of Jupiter diameters in an A.U. (1050 J.D. per A.U.). Enter your converted values in the spaces provided with each graph. You now have all the information you need to use Kepler's Third Law to find the mass of Jupiter. **When recording your calculations, only write down numbers with three significant figures of accuracy. (e.g. 1.23, 12.3, 123, 0.0123)**

$$M = \frac{a^3}{P^2}$$

where M is the mass of the primary body (Jupiter), in units of the solar mass.

a is the radius of the orbit in units 1 A.U.

P is the period of the orbit in Earth years.

Note: If you wish, you can use the Calculator program in Windows 98. It has many scientific functions you might use.